Reliability and Early Longitudinal Change in Infant EEG Flow Geometry

A reanalysis of the APPLESEED example dataset

Article type: Research Article

Short title: Infant EEG flow geometry

Keywords: infant EEG, test--retest reliability, longitudinal development, neural dynamics, Jacobian, reachability

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

Abstract We assessed the reproducibility and early longitudinal change of operational neural-flow summaries in 48 longitudinal EEG recordings from 13 infants. The public APPLESEED example dataset includes two approximately four-month recordings for all children and eight- and twelve-month follow-ups for 11 children. We summarized every recording before inference, treating sessions rather than epochs as the statistical unit. Primary endpoints were three-dimensional Meta-Noetic Jacobian (MNJ) summaries and short-horizon reachability measures computed from subject-anchored coordinates. Test–retest agreement used absolute-agreement intraclass correlations, bootstrap intervals, session differences, and correlation as a descriptive complement. ICC(A,1) values were low or negative (range −0.398 to 0.242). In the balanced 11-child exploratory comparison, the pooled four-month value (mean of sessions 1 and 2) showed lower reachability effective dimension at eight months (paired $d_z = -0.77$, raw p=0.042) and a trend toward higher spectral radius ($d_z = 0.59$, raw p=0.054). These unadjusted results are hypothesis-generating; the available export uses subject anchoring rather than cohort anchoring and does not support absolute between-child coordinate levels, normative trajectories, or latent-mechanism claims.

Author Summary

Infant EEG measures must be stable enough to interpret but sensitive enough to change with development. We use repeated APPLESEED recordings to test whether operational summaries of neural flow and nearby reachable states agree over a short retest interval and how those summaries change within the same children over the first year.

1. Introduction

Repeated EEG measurements are essential for developmental neuroscience, but a feature is useful only if its short-interval variation can be distinguished from measurement instability. APPLESEED was developed to make pediatric EEG preprocessing and scale-wise entropy estimation more reproducible (Puglia et al., 2022). Its example dataset offers a compact test case: two recordings at approximately four months and, for most children, additional recordings at approximately eight and twelve months.

Here we evaluate operational summaries derived from an existing NeuralManifoldDynamics export. Measured EEG is mapped to a three-coordinate chart and a stratified nine-coordinate chart; local flow is then summarized through a Jacobian and through short-horizon reachability. These quantities are measurement-derived summaries rather than direct observations of a latent neural manifold. Their reliability and developmental sensitivity must therefore be tested empirically.

Core question

Do session-level MNJ and reachability summaries show usable short-interval agreement at four months, and how do those same summaries change within children over the first year?

2. Theory

Let $x_t = [m_t, d_t, e_t]^T$ denote the exported three-dimensional measurement chart, whose coordinates operationally summarize mobility, deviation/diffusivity, and entropy-related variation. The local flow is summarized by:

$$J(x,t) = \partial \frac{\dot{x}_t}{\partial} x_t \tag{1}$$

.

We summarize J by its trace, Frobenius norm, antisymmetric rotation norm, and spectral radius. Separately, a local linear model yields a short-horizon covariance recursion:

$$\Sigma_t^{h+1} = A_t \Sigma_t^h A_t^T + Q_t \tag{2}$$

.

Its log-determinant, effective dimension, condition number, directional entropy, and rotation proxy summarize reachable local variation. These are features of the fitted measurement chart and local model, not direct estimates of a latent state.

3. Methods

3.1. Dataset

We analyzed the APPLESEED Example Dataset (OpenNeuro ds003710 version 1.0.2; (OpenNeuro, 2022)), originally accompanying the APPLESEED preprocessing paper (Puglia et al., 2022). The dataset has 48 EEG recordings from 13 infants. All contributed sessions 1 and 2 within approximately one week at four months; 11 children also contributed session 3 at eight months and session 4 at twelve months.

3.2. Export contract

We used the frozen export `neuralmanifolddynamics_ds003710_20260804_123154`. The export contains valid 3D and 9D trajectories, 3D and 9D Jacobians, raw feature summaries, and event-to-window mappings for every recording. We checked the geometry contract before analysis. The export supplies `subject_anchored`, but not `cohort_anchored`, coordinates. All claims therefore retain a within-subject interpretation.

3.3. Endpoints and inference

The primary family contains MNJ Frobenius norm, rotation norm, trace, spectral radius, reachability log-determinant, effective dimension, condition number, directional entropy, and reachability rotation. We used the shared `ndt-analysis` reachability implementation with $k = 12$ and horizon $h = 3$.

The statistical unit was a child-session, never an epoch or Jacobian window. Test–retest analysis paired sessions 1 and 2 for all 13 children. We report ICC(A,1), bootstrap intervals, standard error of measurement, minimum detectable change, paired Wilcoxon tests, and paired d_z (Bland & Altman, 1986; Shrout & Fleiss, 1979). The highlighted exploratory age comparison uses only the 11 children with eight-month data: its four-month value is the mean of sessions 1 and 2 and its eight-month value is session 3. We report paired Wilcoxon raw p-values, paired d_z, and child-cluster bootstrap intervals. Benjamini–Hochberg values are retained in the machine-readable outputs and supplement as a transparency sensitivity analysis, but do not define the exploratory interpretation (Benjamini & Hochberg, 1995).

3.4. Robustness and exploratory analysis

We checked H5 availability, subject/session counts, geometry-contract status, retained Jacobian windows, and finite primary values. Bootstrap intervals resample children, not epochs. A retest-mean sensitivity analysis compares sessions 3 and 4 against the mean of sessions 1 and 2. Event-locked object, water, speech, and face contrasts are exploratory and not used to support the primary claims.

4. Results

4.1. R1: Coverage and validity

The final run included 48 H5 recordings: 13 children at sessions 1–2 and 11 children at sessions 3–4. The source manifest records 3D/9D coordinate and Jacobian availability for all 48 recordings. Coverage and geometry-contract audit results are reported in S3.

Session	Children	Nominal age	Analytic role
1	13	4 months	Reliability baseline
2	13	4 months	Retest
3	11	8 months	Longitudinal follow-up
4	11	12 months	Longitudinal follow-up

Table 1: APPLESEED longitudinal design. The two children without sessions 3–4 remain in the reliability analysis.

4.2. R2: Test–retest reliability

All nine ICC(A,1) estimates were low or negative (range −0.398 to 0.242), with wide child-bootstrap intervals. The maximum estimate was 0.242 for reachability effective dimension (95% interval −0.324 to 0.636); no session 1–2 paired difference survived FDR correction (all adjusted p >= 0.895). Complete estimates are in S2. Absolute coordinate positions are not interpreted because the available coordinates are subject anchored.

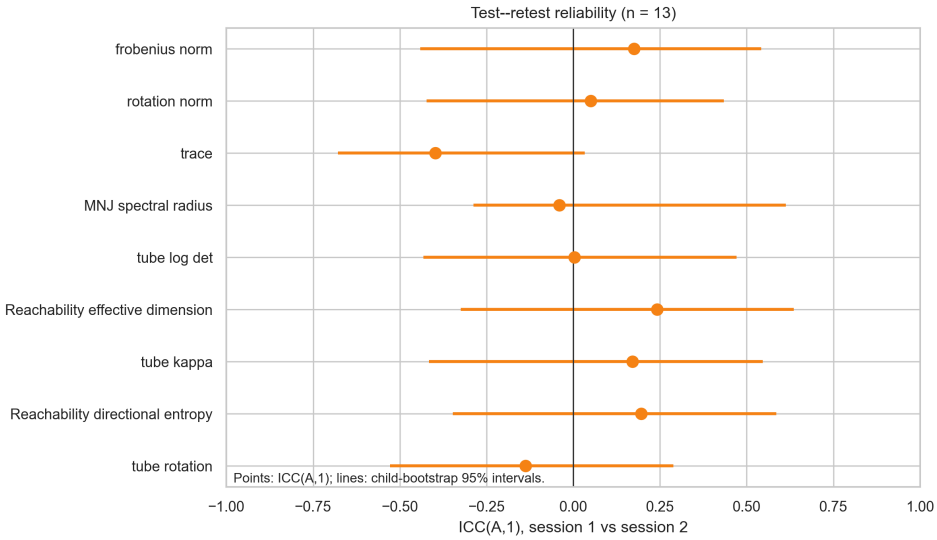


Figure 1: Absolute-agreement test–retest ICCs with child-bootstrap 95% intervals. The broad intervals and low point estimates preclude a strong reliability claim in this example sample.

4.3. R3: Exploratory 4-to-8-month developmental comparison

We restricted this analysis to the 11 children with both four- and eight-month recordings, excluding the two reliability-only children. For each child, the four-month value was the mean of sessions 1 and 2. Reachability effective dimension was lower at eight months (mean difference −0.145, paired $d_z = -0.77$, raw p=0.042; 95% child-bootstrap interval −0.301 to −0.033). Spectral radius was higher at eight months (mean difference 0.00503, $d_z = 0.59$, raw p=0.054). Directional entropy also decreased (mean difference −0.0595, $d_z = -0.66$, raw p=0.102). These are the largest exploratory age-associated effects in the preselected family. They are not multiplicity-corrected discoveries and should be tested prospectively in a larger repeated-measures cohort.

Endpoint	4-month to 8-month difference	Paired d_z	Raw p
`tube_d_eff`	−0.145	−0.77	0.042
`spectral_radius`	0.00503	0.59	0.054
`tube_dir_entropy`	−0.0595	−0.66	0.102

Table 2: Largest exploratory 4-to-8-month effects. Four-month values are each child's mean of sessions 1 and 2; all tests use the same 11 children. Raw p-values are descriptive, not multiplicity-corrected confirmatory tests.

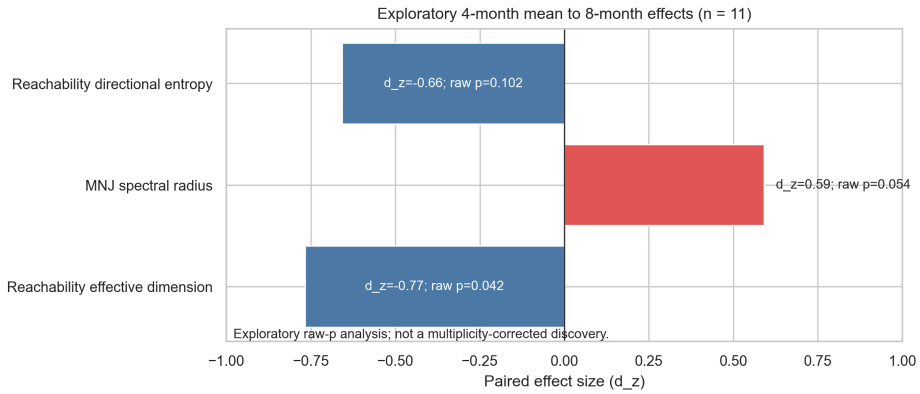


Figure 2: Exploratory paired effect sizes for the three highlighted 4-to-8-month contrasts. Labels give raw p-values; neither color nor annotation denotes multiplicity-corrected significance.

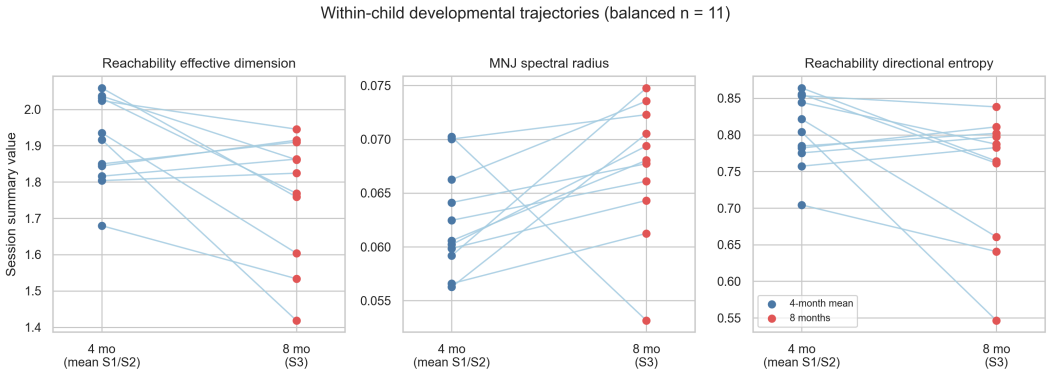


Figure 3: Within-child 4-to-8-month trajectories for the same three exploratory endpoints. Each line is one child; the four-month value is the mean of sessions 1 and 2.

4.4. R4: Robustness and event exploration

All primary analyses aggregate epochs and Jacobian windows before inference. The retest-mean sensitivity analysis and the separate event-window analysis are reported in S3. Neither exploratory result changes the primary endpoint family or its correction scope.

5. Discussion

The contribution of this study is an auditable, session-level evaluation surface for infant EEG flow geometry. Reliability must be interpreted endpoint by endpoint: correlation alone is not absolute agreement, and agreement alone is not developmental sensitivity. Within the balanced 11-child 4-to-8-month comparison, the joint pattern of lower reachability effective dimension and higher spectral radius is consistent with a change in the locally fitted geometry of the exported EEG chart. It is an internal exploratory result, not a validated developmental marker.

MNJ and reachability summaries are local-model outputs on an operational chart. The analysis does not establish that any coordinate is a latent neural variable, that the chart is unique, or that within-child changes are caused by maturation rather than recording-context differences. The small sample, unequal event coverage, locally anchored feature normalization, and absence of cohort anchoring remain material limitations.

This work separates reliability, within-child change, and exploratory event effects in a public longitudinal infant EEG dataset, providing a reproducible template for more highly powered developmental studies.

6. Acknowledgements

We thank the APPLESEED authors and OpenNeuro for releasing the example dataset.

Bibliography

Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Bland, J. M., & Altman, D. G. (1986). Statistical Methods for Assessing Agreement Between Two Methods of Clinical Measurement. *The Lancet*, *327*(8476), 307–310. https://doi.org/10.1016/S0140-6736(86)90837-8

OpenNeuro. (2022,). *APPLESEED Example Dataset*. https://doi.org/10.18112/openneuro.ds003710.v1.0.2

Puglia, M. H., Slobin, J. S., & Williams, C. L. (2022). The Automated Preprocessing Pipe-Line for the Estimation of Scale-Wise Entropy from EEG Data (APPLESEED): Development and Validation for Use in Pediatric Populations. *Developmental Cognitive Neuroscience*, *58*, 101163. https://doi.org/10.1016/j.dcn.2022.101163

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass Correlations: Uses in Assessing Rater Reliability. *Psychological Bulletin*, *86*(2), 420–428. https://doi.org/10.1037/0033-2909.86.2.420

Author Contributions

RL: Conceptualization, Methodology, Data Curation, Analysis, Writing, and Editing.

Funding Information

No dedicated funding supported this secondary analysis.

Data Availability Statement

The APPLESEED example dataset is publicly available through OpenNeuro (ds003710, version 1.0.2; (OpenNeuro, 2022)). Analysis wrappers and derived tables are contained in this repository.

Competing Interests

The author declares no competing interests.